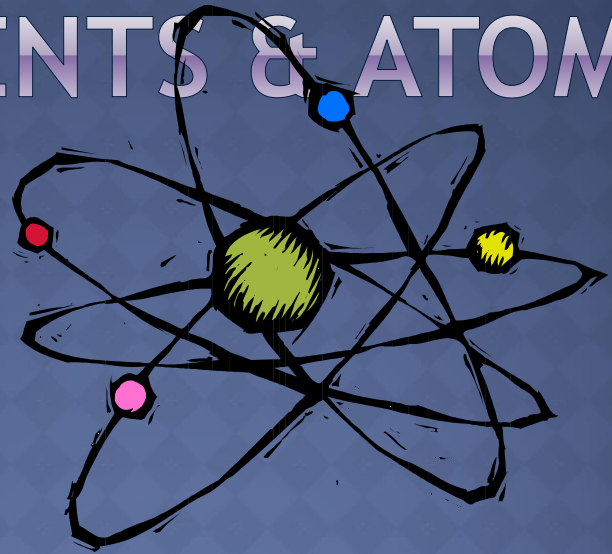


ELEMENTS & ATOMS

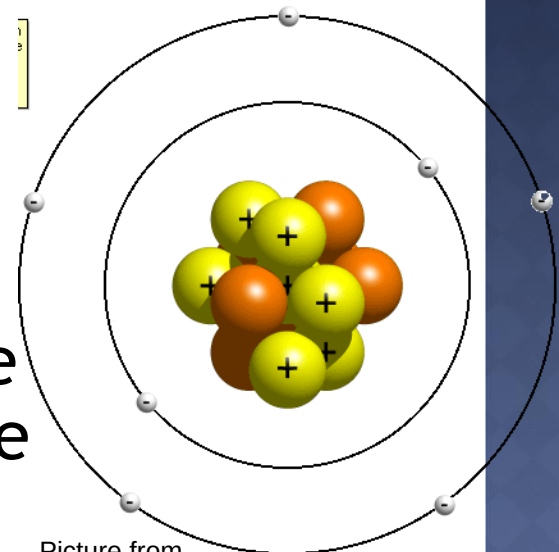


AN ATOM REFRESHER

- ◉ **Matter** is anything that takes up space and has mass.
- ◉ All matter is made of atoms.
- ◉ Atoms are the building blocks of matter, sort of how bricks are the building blocks of houses.

AN ATOM REFRESHER

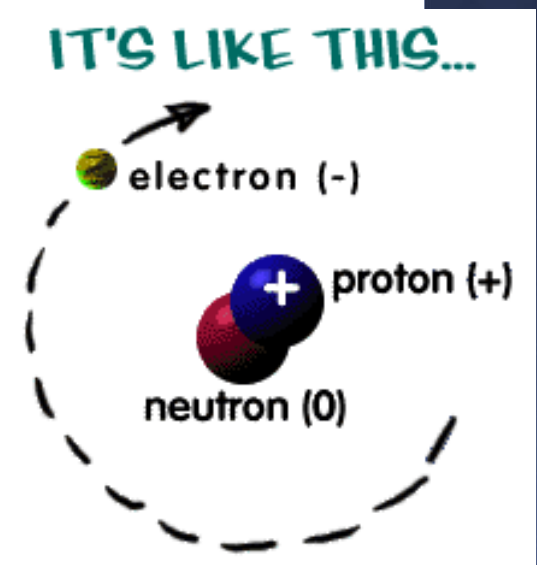
- An atom has three parts:
 - Proton = positive
 - Neutron = no charge
 - Electron = negative
-
- The proton & neutron are found in the center of the atom, a place called the nucleus.
 - The electrons orbit the nucleus.



Picture from
http://education.jlab.org/qa/atom_model_03.gif

WHAT ARE ELEMENTS?

- ◉ Elements are the alphabet to the language of molecules.
- ◉ To make molecules, you must have elements.
- ◉ Elements are made of atoms. While the atoms may have different weights and organization, they are all built in the same way.



Atoms always have as many electrons as protons.
Atoms usually have about as many neutrons as protons.

Hydrogen



1 proton
1 electron
0 neutrons

Helium



2 protons
2 electrons
2 neutrons

Carbon

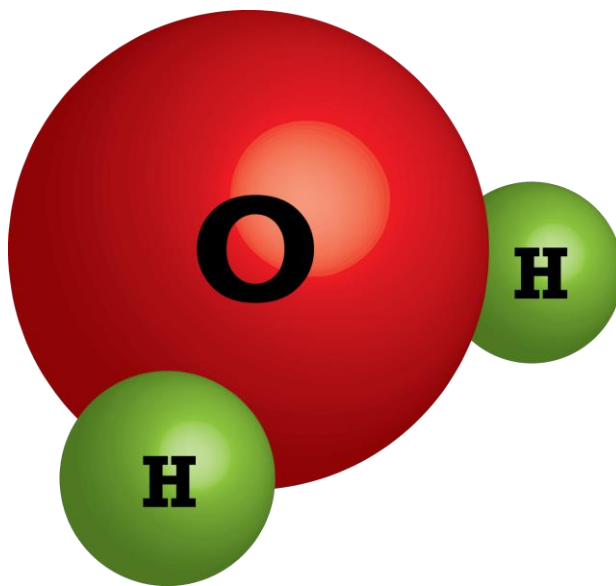


6 protons
6 electrons
6 neutrons

Adding a proton makes a new kind of atom!
Adding a neutron makes an isotope of that atom,
a heavier version of that atom!

WHAT ARE COMPOUNDS

- Compounds are composed of two or more elements chemically joined in a fixed ratio.



MORE ABOUT ELEMENTS..

- ◉ **Elements** are the building blocks of all matter.
- ◉ The periodic table is a list of all of the elements that can build matter. It's a little like the alphabet of chemistry.



Writing the symbols for the elements..

- The symbol is always printed e.g. Al not al
- If an element's symbol has more than one letter, the first letter must be in CAPITALS and the others in lower case.

ELEMENTS

- ◉ You should learn the names and symbols for the following elements:
 - First 20 elements
 - Common elements: Cobalt, Copper, Gold, Lead, Mercury, Nickel, Silver, Iron, Uranium, Tin, Zinc

Atomic Number:

Number of protons
and it is also the
number of electrons
in an atom of an
element.

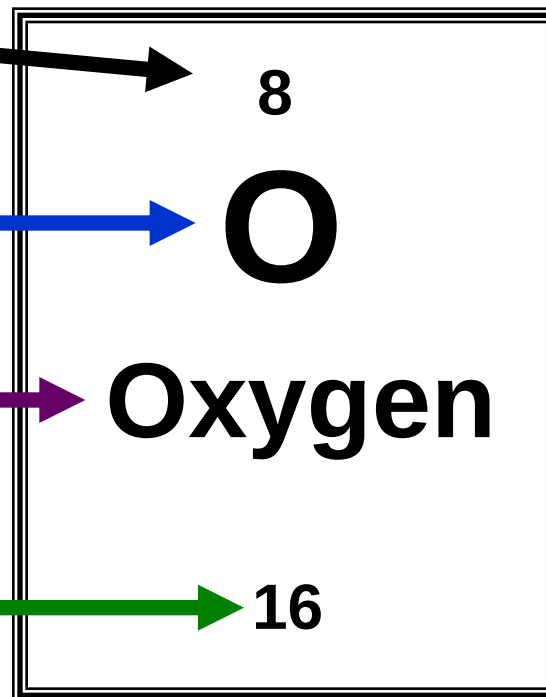
Element's Symbol:

An abbreviation for
the element.

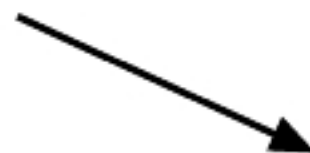
Elements Name

Atomic Mass/Weight:

Number of protons +
neutrons.



Mass number (A)



23

11

Na

Atomic number (Z)

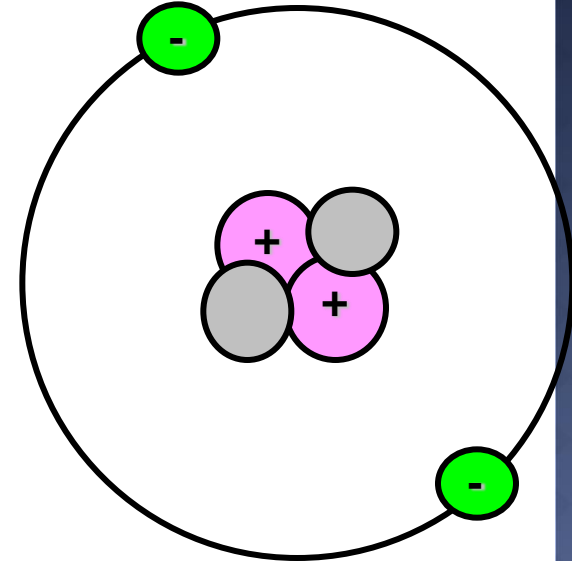


ATOM MODELS

- ⦿ There are two models of the atoms we will be using in class.
- ⦿ Bohr Model
- ⦿ Lewis Dot Structure

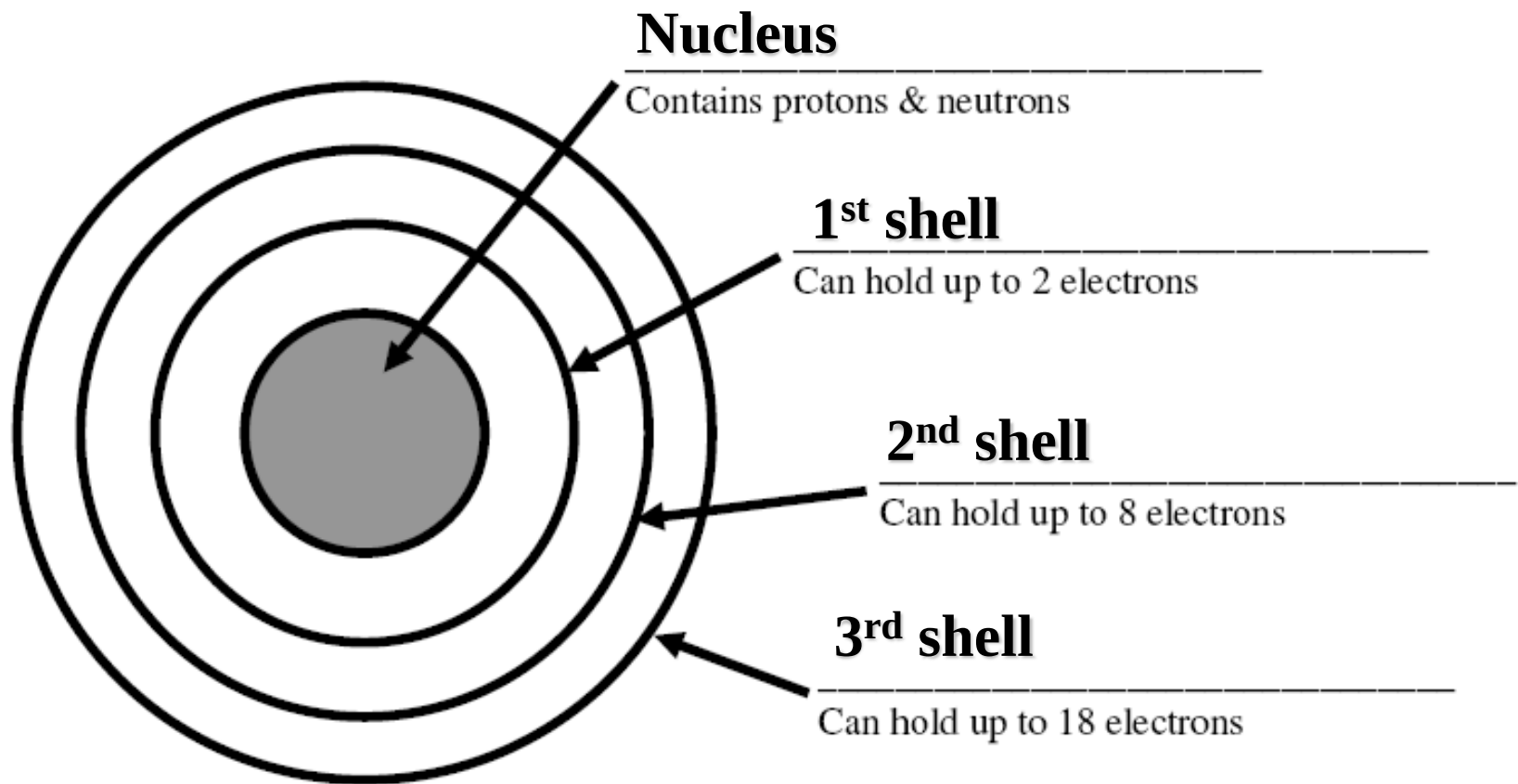
BOHR MODEL

- The Bohr Model shows all of the particles in the atom.
- In the center is circles. Each circle represents a single neutron or proton. Protons should have a plus or P written on them. Neutrons should be blank or have an N.
- In a circle around the nucleus are the electrons. Electrons should have a minus sign or an e.



ELECTRONS HAVE SPECIAL RULES....

- ⦿ You can't just shove all of the electrons into the first orbit of an atom.
- ⦿ Electrons live in something called shells or energy levels.
- ⦿ Only so many can be in any certain shell.

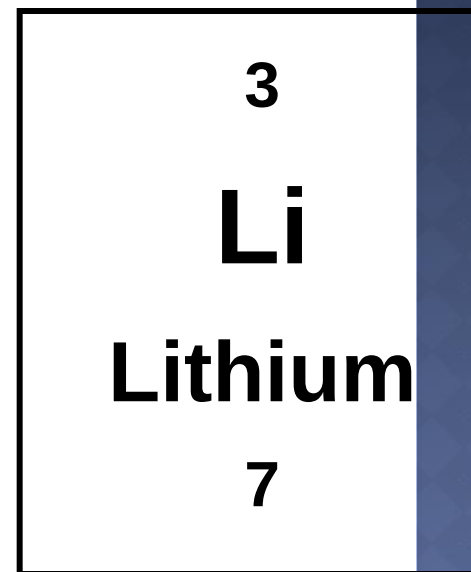


ELECTRONS HAVE SPECIAL RULES....

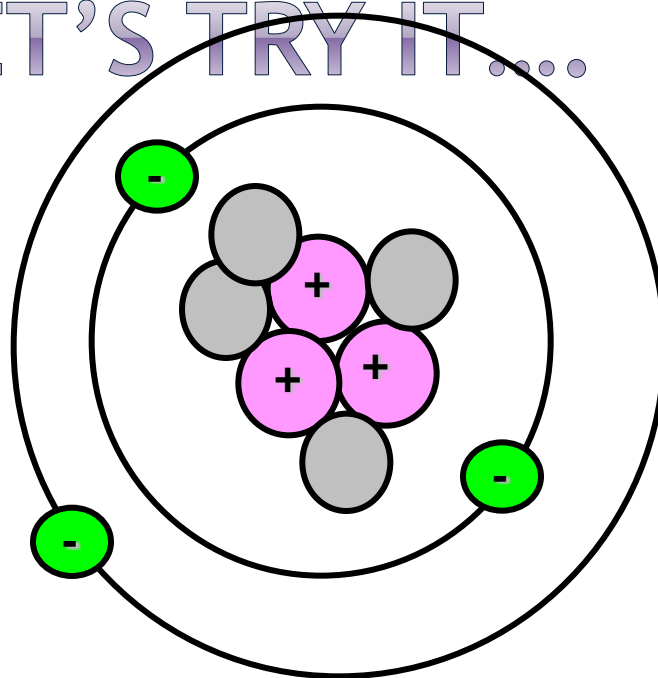
- ⦿ You can't just shove all of the electrons into the first orbit of an electron.
- ⦿ Electrons live in something called shells or energy levels.
- ⦿ Only so many can be in any certain shell.
- ⦿ The electrons in the outer most shell of any element are called valance electrons.

SO LET'S TRY IT....

- How to draw a Lithium atom
- First, look at the Periodic Table
- Second, determine the number of protons (look @ the atomic number)
- Then determine the number of neutrons (atomic mass - atomic number)
- Then determine the number of electrons (look @ the atomic number)



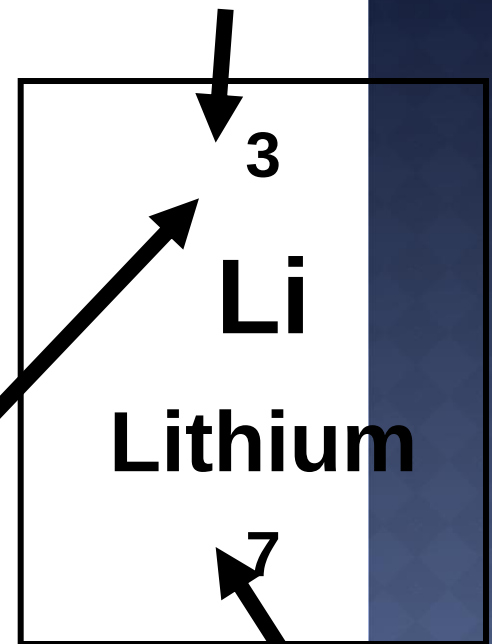
SO LET'S TRY IT....



Electrons = 3

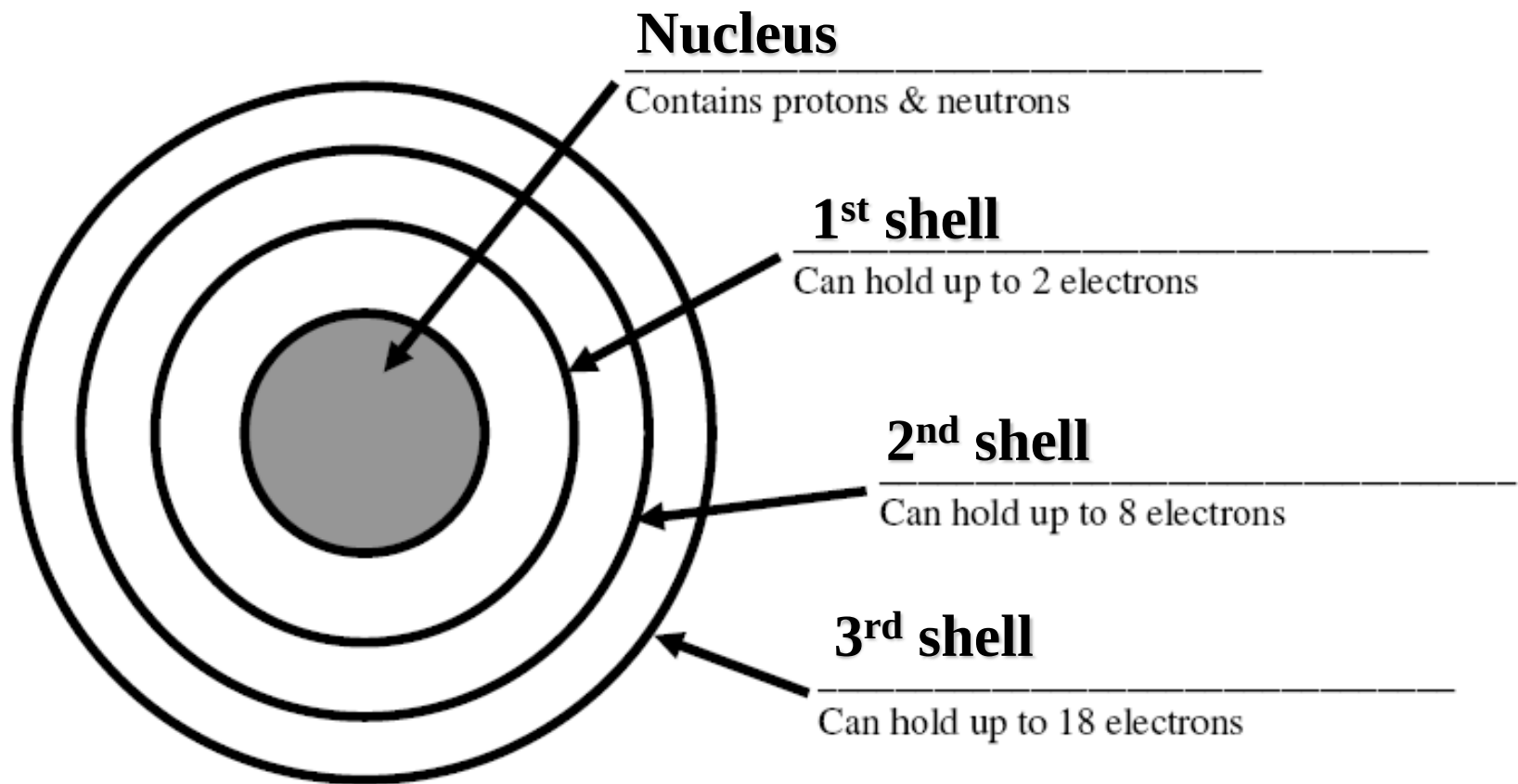
2 in the 1st shell, 1 in the 2nd shell

Protons = 3



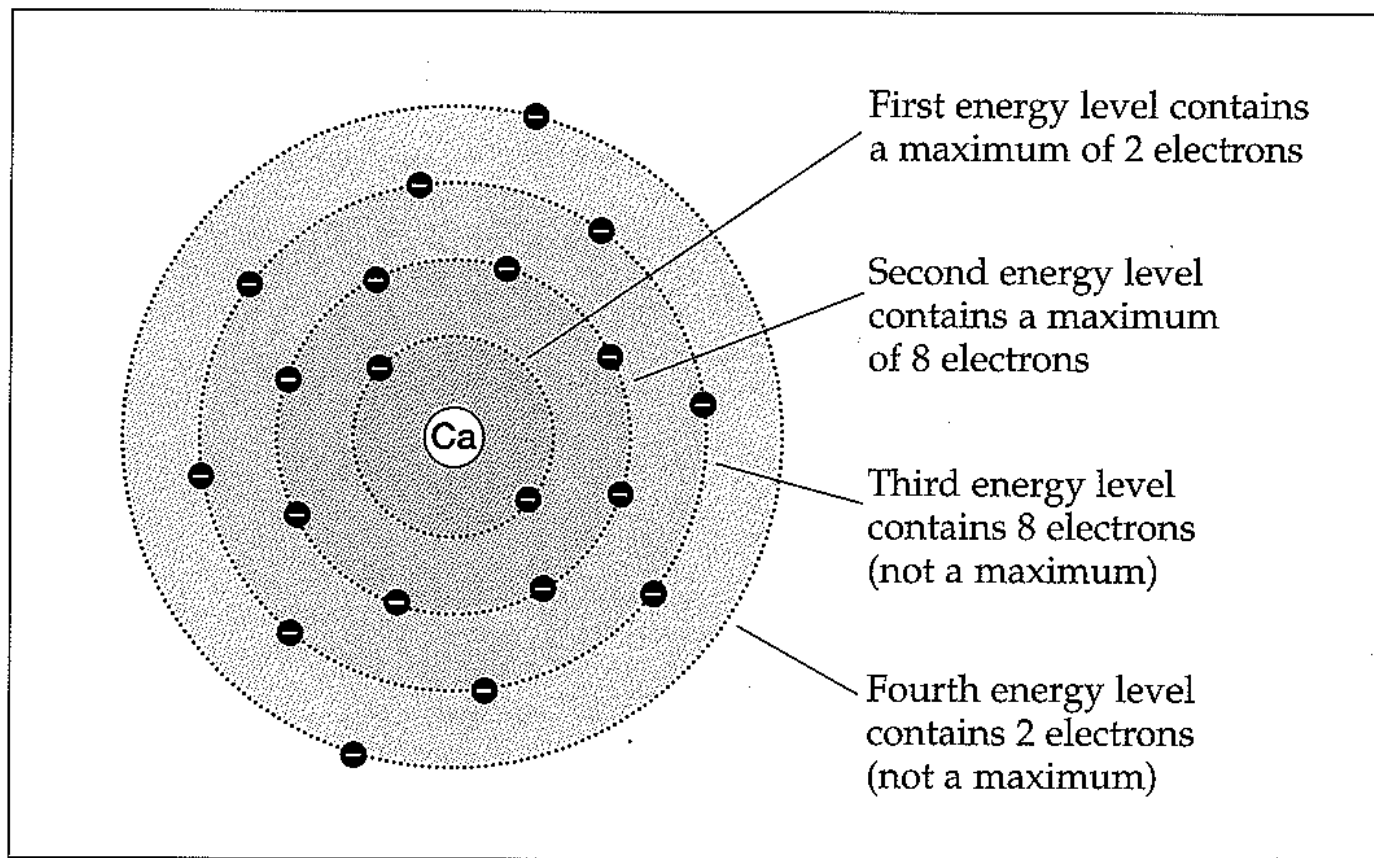
Neutrons = 4

(7-3=4)



Although they cannot say exactly where an electron is and how fast it is travelling, scientists can determine the arrangement of electrons around the nucleus.

A simple model that shows how electrons exist around the atom is shown below for an atom of calcium. The electrons occupy **ENERGY LEVELS**.



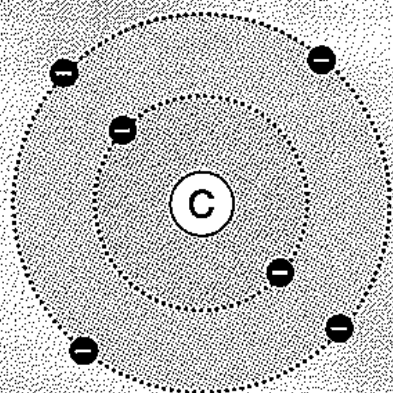
An abbreviated way of writing this electron structure for calcium would be:

Ca 2 , 8 , 8 , 2

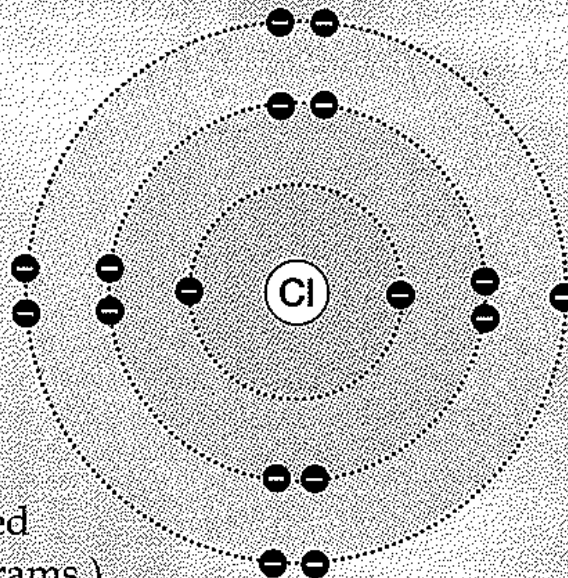
This is called its **ELECTRON CONFIGURATION**.

Example 2.1 : Show the arrangement of electrons around:

a) carbon (${}^{12}_6\text{C}$)



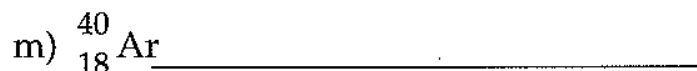
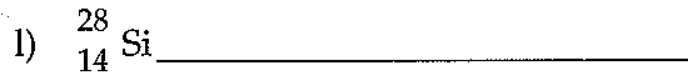
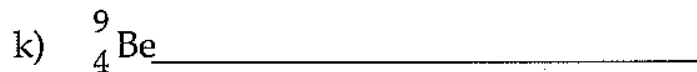
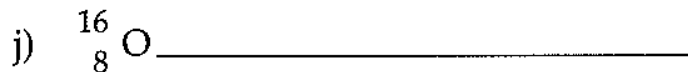
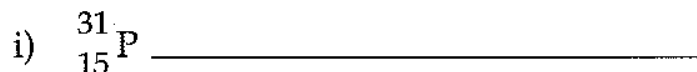
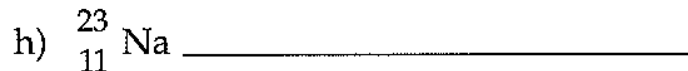
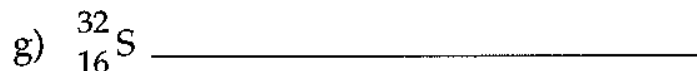
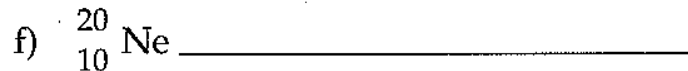
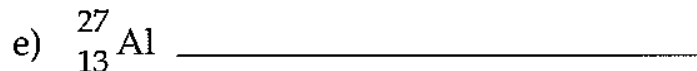
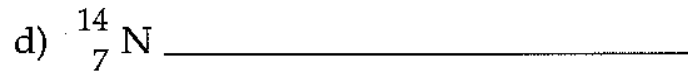
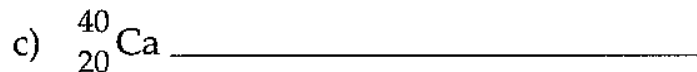
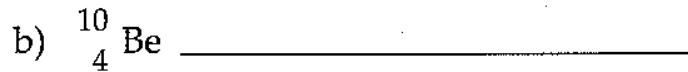
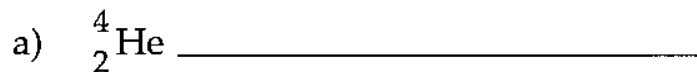
b) chlorine (${}^{35}_{17}\text{Cl}$)



(These are sometimes called
electron energy level diagrams.)

CHECKPOINT!

Question 2.2 - Write the electron configuration for each of the following elements:



REVIEW

Draw the electron energy level diagrams and state the electron configuration for each of the following elements: F, K, H, Si, Mg, Ar